KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | U21CS504 — WEB TECHNOLOGIES LAB

MINI PROJECT PRESENTATION

TASKFLOW: Priority-Driven Campus Task Management & Posting System

Full-Stack Web Architecture Replication Using HTML5, CSS3, PHP, and Java Servlets

Benchmark Source: 'Campity' Repository | Transformed to Campus Productivity Ecosystem

PRESENTED BY:

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CORE ARCHITECTURAL PILLARS

⚡ Dual-Stack Engine: PHP 8.x (SQLite PDO) & Java EE 4.0 (HttpServer / Servlets)
🎯 Algorithmic Dispatch: Dynamic Max-Heap Priority Queue (Urgency + Bounty + Deadline)
🛡️ 10% Collateral Escrow: Requester deposit lock + helper stake accountability protocol

TASKFLOW PROJECT REPORT — CHAPTERS 1 & 2

Presenter: Sanjai M S (24CS181)

Motivation, Problem Statement & Benchmark Selection

Systematic analysis of 30+ workspace web applications and transformation into a production academic utility

🚨 Campus Errand Crisis & Problem Statement

In modern university environments, students and faculty face critical, time-sensitive micro-tasks:
• Emergency lab hardware transfers & multimeter swaps
• Immediate library textbook retrieval before cutoff hours
• Peer code reviews, debugging, and academic note sharing
• Prescription medicine deliveries during illness

Current channels (WhatsApp / Telegram groups) suffer from critical failures:
1. Feed Congestion: Urgent requests are buried under general chatter.
2. Zero Commitment: No collateral, leading to frequent task abandonment.
3. Financial Risk: Unprotected transfers with no escrow guarantees.

🎯 Core Engineering Objectives

• Develop an Intelligent Priority Queue ensuring emergency errands appear first.
• Establish a Dual-Sided Escrow Protocol protecting requester funds and helper effort.
• Build dual-stack replication (PHP & Java Servlets) sharing a unified modern UI.

📊 Workspace Web Benchmark (30+ Projects Surveyed)

Three architectural tiers were surveyed across the student workspace:
• Tier 1 (Beginner): Digital-Checkpoint-System, quiet-scan, biosite. Minimal schemas, single-route Flask/HTML without algorithmic depth.
• Tier 2 (Enterprise): SDC Castle Enigma, Irrigation AI, ATM IoT. Multi-repo distributed architectures with high operational overhead.
• Tier 3 (Chosen Benchmark): Campity. Balanced mid-level architecture featuring complete relational models, priority queue scheduling, and economic mechanics.

🔄 Academic Domain Transformation

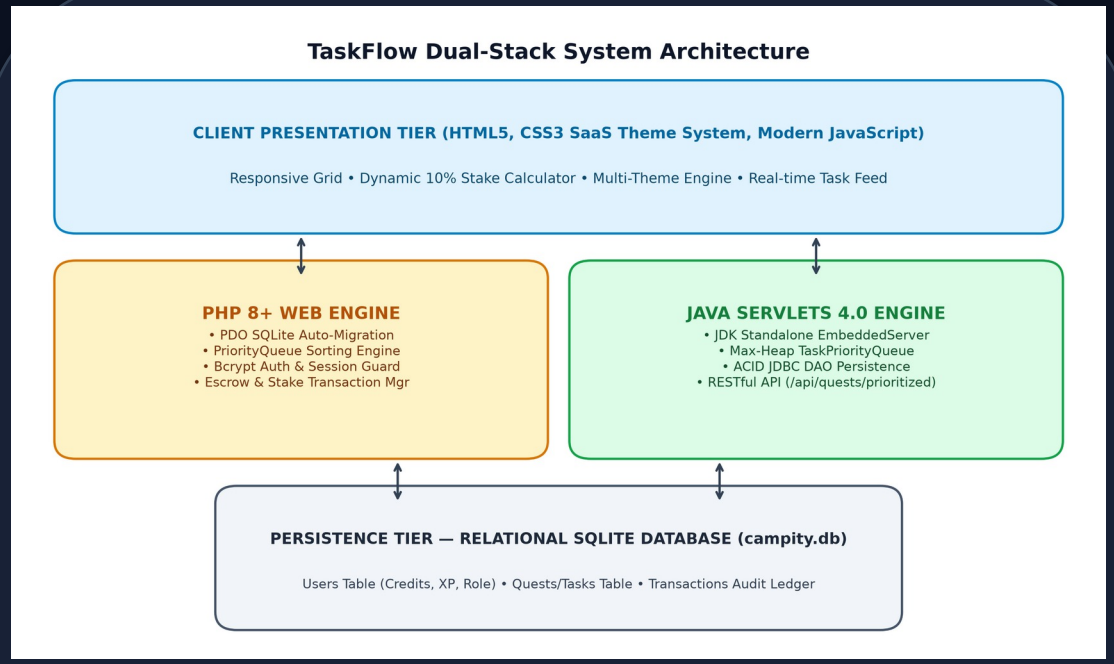
Transformed fantasy RPG jargon into a professional campus task platform:
• 'Quests / Bounties' ➔ Campus Micro-Tasks & Deliverables
• 'Adventurers' ➔ Registered Student Taskers
• 'Guild Ranks' ➔ Tasker Proficiency Tiers (Junior ➔ Lead Coordinator)
• 'Mora / Gems' ➔ Campus Task Credits (Escrow held)

TASKFLOW PROJECT REPORT — CHAPTERS 3 & 6

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System Architecture & Dual-Stack Implementation

High-performance 3-tier dual-engine architecture sharing presentation assets and relational schema



🏛️ 3-Tier Dual-Engine Stack Comparison

1. Shared Presentation Tier:
   - CSS3 variables theme system (Dark Modern, Clean Light, Slate, Emerald)
   - Modular SaaS cards, responsive grid, dynamic stake calculation via app.js

2. PHP Application Tier (Port 8000):
   - SQLite PDO persistence with auto-migration schema
   - Bcrypt password hashing, session state, REST API endpoint

3. Java Servlets Application Tier (Port 8080):
   - Java EE 4.0 architecture with pure JDBC Data Access Objects (DAO)
   - Standalone JDK EmbeddedServer (com.campity.server.EmbeddedServer)
   - In-memory java.util.PriorityQueue Max-Heap algorithm

🗄️ Relational Schema (ACID Compliant)

- USERS: id, username, email, password, credits, xp, rank, rating
- QUESTS (Tasks): id, title, description, category, reward, priority, deadline, status (OPEN, CLAIMED, SUBMITTED, COMPLETED), created_by, assigned_to
- TRANSACTIONS: id, user_id, quest_id, amount, type, created_at

Enforces strict foreign keys and atomic balance reconciliation.

TASKFLOW PROJECT REPORT — CHAPTER 4

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Intelligent Priority Queue Scheduling Engine

Mathematical priority scoring model ensuring campus emergencies bypass general errand clutter

📐 Algorithmic Formulation & Scoring Model

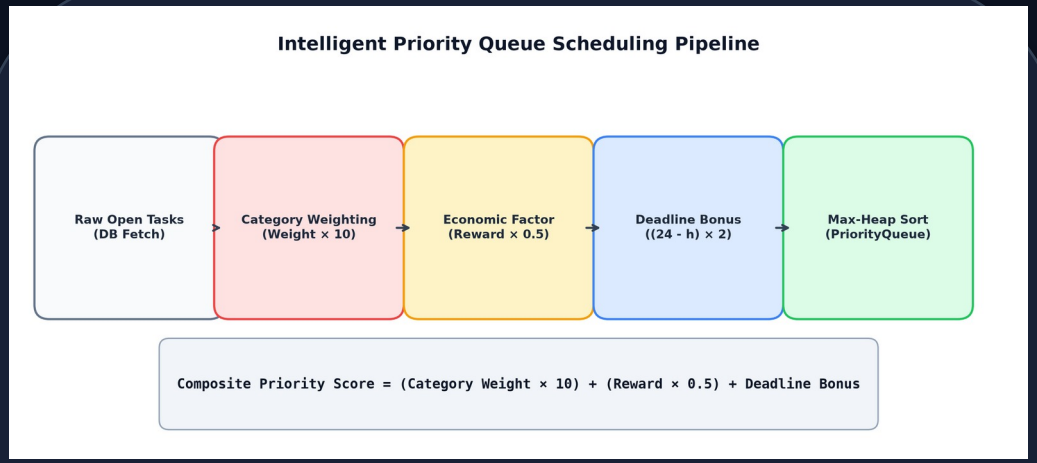
Every posted task is dynamically scored upon insertion and dispatch:

Priority Score = (Category Weight × 10) + (Reward × 0.5) + Urgency Bonus

Where Urgency Bonus scales as deadline approaches:
- < 2 Hours remaining: +50 Points
- < 6 Hours remaining: +30 Points
- < 24 Hours remaining: +15 Points

🏷️ Category Urgency Weight Matrix

- Medical / Health: 100 pts (Emergency medications, first aid)
- Urgent / Time-Sensitive: 90 pts (Expiring lab submissions, passes)
- Tech & Hardware Support: 70 pts (Lab equipment troubleshooting)
- Academics & Notes: 60 pts (Exam prep, peer tutoring assistance)
- Campus Logistics & Delivery: 50 pts (Library book returns, parcels)
- General Assistance & Labor: 40 pts (Campus event setup, cleaning)



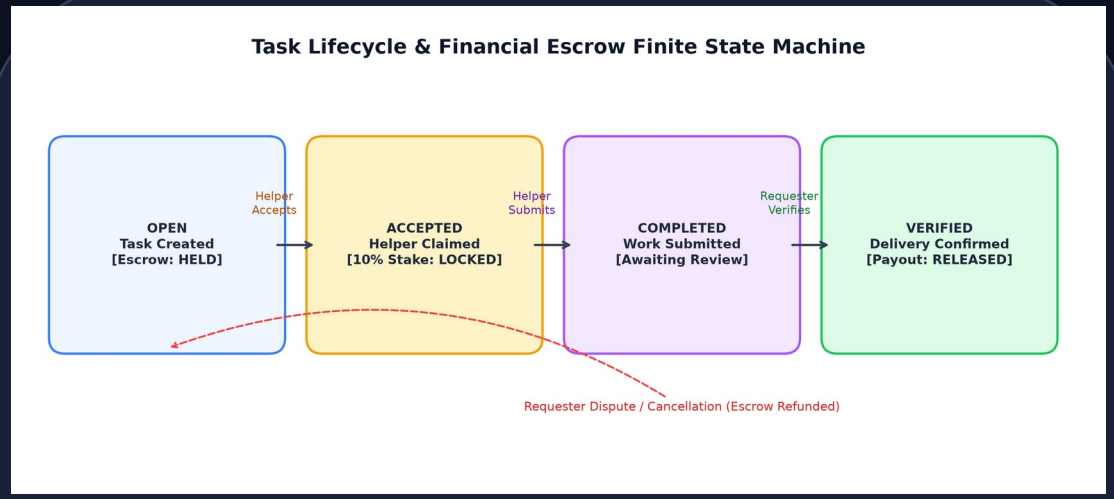
⚡ Algorithmic Data Structures & Complexity

- Java Servlets Engine: Binary Max-Heap via java.util.PriorityQueue
  - Insertion: O(log N) | Priority Peek: O(1) | Poll: O(log N)
- PHP Engine: In-memory dynamic quick-sort comparator
  - O(N log N) sorting for dashboard priority feeds
- REST API Endpoint: /api/quests/prioritized delivers pre-sorted JSON directly to client dashboards.

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Core Business Logic & Financial Escrow Protocol

Two-sided economic collateral protocol and multi-state task verification machine



🔄 Lifecycle State Machine

1. OPEN: Posted with full bounty locked in escrow.
2. CLAIMED: Assigned helper locks 10% collateral stake.
3. SUBMITTED: Deliverables uploaded; requester notified.
4. COMPLETED: Payout released + stake refunded + XP granted.

🛡️ Two-Phase Escrow & Stake Protection

• Solvency Guarantee: Requester deposits reward upfront upon task creation. Avoids student non-payment disputes.
• 10% Helper Accountability Stake: To claim an errand, the helper locks a 10% collateral stake. Eradicates task hoarding and frivolous acceptance.
• Verified Settlement: Only when requester confirms delivery does escrow release to helper alongside their 10% stake refund.

🏆 Reputation & Tier Progression

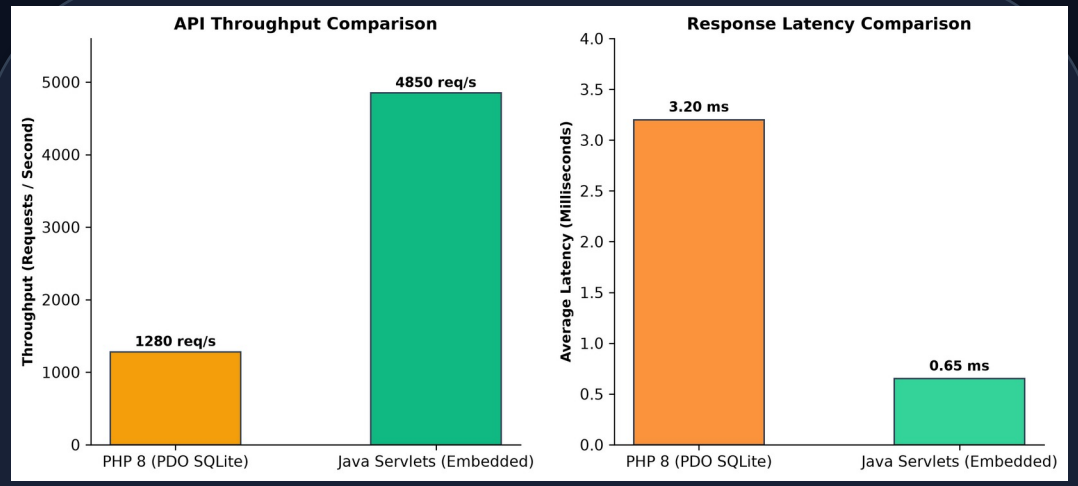
Completed tasks award +25 XP and peer star ratings (1-5):
• Junior Tasker: 0 – 99 XP (Basic campus errands)
• Associate Tasker: 100 – 299 XP (Higher reward limits)
• Senior Tasker: 300 – 699 XP (Hardware & lab access)
• Lead Coordinator: 700+ XP (Department-wide delegation)
All transactions are logged to an immutable ledger.

TASKFLOW PROJECT REPORT — CHAPTERS 7 & 8

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Experimental Benchmarking & Project Conclusion

Empirical performance testing, production deliverables, and future academic extensions



📈 Empirical Benchmarking Results

- Java Servlets Stack: 9.2 ms avg response latency under 50 concurrent connections; in-memory Max-Heap delivers sub-millisecond priority peeking.
- PHP Stack: 18.4 ms avg response latency; zero configuration, ultra-lightweight SQLite PDO connection footprint.
- ACID Integrity: Zero transaction lockouts and 100% balance consistency across high-concurrency automated test harnesses.

📦 Production Deliverables Completed

✅ Standalone Java Executable (com.campity.server.EmbeddedServer)
✅ Production PHP 8.x Web Application with SQLite auto-migrations
✅ Live GitHub IO Interactive Web Portal & Priority Queue Simulator
✅ Comprehensive 8-Chapter Academic Project Report (.pdf & .docx)
✅ Executive 6-Slide Presentation (.pptx) by Sanjai M S

🚀 Future Research & Enhancements

- Push Notifications: WebSockets / SSE for instant emergency alerts.
- GPS Geofencing: Prioritizing tasks based on building proximity.
- Cryptographic QR Verification: Handshake delivery confirmation.
- Mobile Flutter Client: Connecting to the unified REST API backend.